Lec-14

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int [] arr = new int [5]

① ArrayList → Class → NP → Heap

```java
public static void main(String[] args) {
    // TODO Auto-generated method stub
    ArrayList<Integer> ll = new ArrayList<>();
    // add
    ll.add(10);// O(1)
    ll.add(20);
    ll.add(30);
    ll.add(40);
    System.out.println(ll);
}
```

| 10 | 20 | 30 | 40 |

0, 9 → 9 10 20 30 40

1 → 10 9 20 30 40

2 → 10 20 9 30 40

3 → 10 20 30 9 40

4 → 10 20 30 40 9

```java
public static void main(String[] args) {
    // TODO Auto-generated method stub
    ArrayList<Integer> ll = new ArrayList<>();
    // add
    ll.add(10);
    ll.add(20);
    ll.add(30);
    ll.add(40);
    ll.add(-9);
    ll.add(7);
    ll.add(8);
    ll.add(9);
    ll.add(1);
    ll.add(15);
    System.out.println(ll.size());
}
```

ll.add(21)

| 10 | 20 | 30 | 40 | -9 | 7 | 8 | 9 | 1 | 15 |

10 × 1.5 = 15

15 × 1.5 = 22.5

23

| 10 | 20 | 30 | 40 | -9 | 7 | 8 | 9 | 1 | 15 | 21 | | | | |

×2

1 | 10 |

2 | 10 | 20 |

30 | 10 | 20 | 30 | 40 |

40

50 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 |

1

1+1 = 2

2+1 = 3

4+1 = 5

8 → 15

1 → 15/8 = 2

vector

Arrays-Sum Of Two Arrays

Take as input N, the size of array. Take N more inputs and store that in an array. Take as input M, the size of second array and take M more inputs and store that in second array. Write a function that returns the sum of two arrays. Print the value returned.

Example 1

Input

```
4
1 0 2 9
5
3 4 5 6 7
```

Output

```
3, 5, 5, 9, 6, END
```

Explanation

```
Sum of [1, 0, 2, 9] and [3, 4, 5, 6, 7] is [3, 5, 5, 9, 6] and the first
digit represents carry over , if any (0 here ) .
```

0 0 1
1 0 2 9
3 4 5 6 7
3 5 5 9 6

18 digit

9 % 10

carry = 8 1

carry = 16 / 10

9 / 10 = 0

0 0 1 0 2 9
3 2 4 5 6 7
3 2 5 5 9 6

sum = carry + 9 + 7

= 16 % 10

1 + 2 + 6 = 9

1
1 7 7
9 8
1 7 5

carry = 1 0

0
2 2
3 5
6 7

n, m

12

| 2 | 3 | 6 | 6 | 5 | 6 |

2 3 4 5
6 5 4 2 8 7
6 5 6 6 3 2

```java
public static void Sum_Of_Two_Array(int[] arr1, int[] arr2) {
        ArrayList<Integer> ll = new ArrayList<>();
        int i = arr1.length - 1;
        int j = arr2.length - 1;
        int carry = 0;
        while (i >= 0 && j >= 0) {
            int sum = carry + arr1[i] + arr2[j];
            ll.add(sum % 10);
            carry = sum / 10;
            i--;
            j--;
        }
}
```

[2, 3, 4]

[5, 7, 9, 3, 4]

carry = 0 1

| 8 | 6 | 1 |

```java
public static void Sum_Of_Two_Array(int[] arr1, int[] arr2) {
        ArrayList<Integer> ll = new ArrayList<>();
        int i = arr1.length - 1;
        int j = arr2.length - 1;
        int carry = 0;
        while (i >= 0 && j >= 0) {
            int sum = carry + arr1[i] + arr2[j];
            ll.add(sum % 10);
            carry = sum / 10;
            i--;
            j--;
        }
        while (i >= 0) {
            int sum = carry + arr1[i];
            ll.add(sum % 10);
            carry = sum / 10;
            i--;

        }
        while (j >= 0) {
            int sum = carry  + arr2[j];
            ll.add(sum % 10);
            carry = sum / 10;
            j--;
        }

    }
```

carry = 0 1

i<0

| 7 | 8 |

| 9 | 8 | 8 |

= 0 + 8 + 8

= 1 + 7 + 8

1 + 9 = 10

| 6 | 6 | 0 | 1 |